3. A hashing function, RIPEMD160, is applied to the public key to obtain the account address:
    a. This address is shorter than the key: 160 bits or 20 bytes. This address is the account number you see in Remix and the Ganache environment and can be used as an address on a public blockchain network, as you'll do later in this chapter (section 5.2.3).
    b. The address is represented in hexadecimal for easy readability, as indicated by the 0x as the first two characters, as in the example 0xca35b7d915458ef540 ade6068dfe2f44e8fa733c.

You've been using EOA addresses to send messages to smart contracts, store ether, and transact on the blockchain. For obvious reasons, account addresses have the stringent requirement to be universally unique. This crucial requirement is addressed by choosing a large (256-bit) address space and by using a cryptographic mechanism to generate collision-free (unique) addresses.

5.2.2 Transaction signing

The cryptographic key pair is also used for Tx signing. The private key is used in the process of digitally signing transactions for authorization and authentication. Recall that in chapter 4, you used MetaMask to confirm your Txs. One of the operations performed at that time by MetaMask is signing the Txs by using your private key. Similar to how you secure and protect a credit card, you need to protect your private key to ensure the security of your assets on the blockchain. Thus, the two main applications of cryptography besides encryption are generating account addresses (the identity of decentralized participants and entities) and digitally signing transactions and messages. Let's apply these two concepts in deploying a smart contract on a public chain for the first time in this book.

5.2.3 Deploying smart contracts on Ropsten